A Unified LLM-Based Framework for Resume–Job Description Semantic Alignment Using PuterJS

Leveraging a Serverless Architecture for Consistent and Accessible ATS Evaluation

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Abstract. Applicant Tracking Systems (ATS) are now a standard part of recruitment processes, posing a significant barrier to job seekers, especially graduates and entry-level workers. Most commercial resume analysis tools and services charge high fees, have very different scoring criteria, and give no clear indication of what criteria they are using. In this paper, we describe Lumen, a free, open source, cloud-based resume analysis tool that provides structured, job description aware, feedback about the resumes it gets. It is built on top of the Puter.js cloud-native platform and is driven by Claude Sonnet 4 large language model. In contrast to other ATS scoring tools, Lumen uses a completely backend API and key free serverless frontend-only architecture to make a feature only previously available to the employers that can afford it, available to all for free. The tool rates the resumes across 5 criteria including ATS friendliness, language use, content, structure, and skill level and gives relevant, job description specific, feedback. Our initial trials of the web app have shown that it can deliver a consistent level of high quality, granular, feedback about the suitability of the resume to a given job, while costing the job seeker nothing at all. This work adds to a body of research that is building the case for using AI in career development and also provides a concrete implementation pattern for building completely free, AI-based, applications that do not intrude on users' privacy using contemporary cloud-native frontend technologies.

Keywords: Applicant Tracking System · Resume Analysis · Artificial Intelligence · Large Language Models · Serverless Architecture · Puter.js · Claude Sonnet 4 · NLP · Career Services · Cloud Computing

1 Introduction

1.1 Background

The modern recruitment landscape has changed with the rise of Applicant Tracking Systems (ATS). Jobscan's 2025 report [1] shows that 97.8% of Fortune 500 companies use an ATS. Workday alone powers over 39% of these systems. The tracking system ranks candidates based on keyword matching, formatting, and job description [5]. This means that the success of the candidate largely hinges on how well their resume showcases their qualifications in a format that machines can read.

This situation unevenly impacts fresh/recent graduates and those changing careers. Fuller et al. [3] estimated that automated filters leave out more than 27 million qualified workers. Cappelli [2] documented the contradiction of employers struggling to find talent while their ATS reject most applicants. Raghavan et al. [4] revealed major gaps in how algorithmic hiring vendors handle bias. At the same time, improvements in LLMs—from GPT-2 [10] to InstructGPT [12] and Claude 3 [13] have led to systems that can understand [14] documents better. This opens new possibilities for making resume evaluations more accessible.

1.2 The Problem

Existing resume analysis platforms (Jobscan, Resume Worded, Kickresume, Zety, etc) have four main limitations:
(1) **Cost barriers**: full analysis requires subscriptions of ₹1,500–6,000+ INR/month;
(2) **Scoring inconsistency**: the same resume gets very different scores on different platforms;
(3) **Generic feedback**: most tools ignore the specific job description;
(4) **Privacy concerns**: sensitive data is unclear to opaque third-party servers.

1.3 Solution: How do we address this?

We present an AI-powered resume analysis tool, **Lumen**, with the following key contributions:

- Zero-cost, fully-functional analysis with no feature gating or limits.
- Job description-aware analysis using Claude Sonnet 4 [13].
- Serverless, backend-free design with Puter.js [20], with auth, storage, and AI services integrated from the frontend [18, 19].
- Multi-dimensional analysis across five different aspects–(ATS compatibility, tone, content, structure, skills).
- Reproducible analysis with a well-defined prompt [16] that generates structured JSON output.

2 Literature Review

2.1 Applicant Tracking Systems: History and Impact

Applicant Tracking Systems were initially developed in the late 1990s as database-oriented applications, which were aimed at processing large volumes of applications. Yu et al. [5] described the first known algorithmic solution for information extraction from resumes, using a cascaded hybrid approach that combined Hidden Markov Models with rule-based methods in their 43rd ACL paper. Today, ATS solutions utilize much more advanced methods, including semantic parsing and ML-based ranking. In their 2025 ATS Usage Report [1], Jobscan discovered that Workday is used in over 39% of Fortune 500 companies alone.

This paradox of employer difficulties in qualified applicant sourcing despite the use of ATS filters that eliminate the majority of applicants was first identified by Cappelli [2]. The hidden workforce issue was quantified by Fuller et al. [3], who estimated over 27 million workers are eliminated from consideration. Raghavan et al. [4] continued to examine the claims of bias mitigation by vendors of algorithmic hiring tools.

2.2 AI and NLP in Resume Screening

The Transformer architecture proposed by Vaswani et al. [6] in their paper "Attention is all you need" replaced traditional recurrent and convolutional architectures with self-attention mechanisms. Devlin et al. [7] proposed another pre-trained deep bidirectional transformer architecture called BERT, which attained state-of-the-art results on eleven NLP tasks.

Roy et al. [8] proposed a resume recommendation system using machine learning. They showed that the Linear SVM classifier was best suited for their application. Jiechieu and Tsopze [9] proposed a CNN-based multi-label classification approach for skill prediction from resumes. The proposed approach attained 98.79% recall and 91.34% precision on 30,000 IT resumes.

2.3 Large Language Models and Their Evolution

The ability of language models to perform zero-shot NLP tasks has been demonstrated by Radford et al. [10] using GPT-2 with 1.5B parameters. Brown et al. [11] extended this to GPT-3 with 175B parameters, confirming that few-shot prompting without gradient updates is sufficient to obtain competitive results. Ouyang et al. [12] presented InstructGPT, confirming that alignment using RLHF makes a 1.3B parameter model preferred to a 175B parameter unaligned model. The release of Claude 3 by Anthropic [13] has provided multimodal capabilities and advanced reasoning. The survey by Zhao et al. [14] is a wide-ranging survey of the LLM space.

2.4 Existing Resume Analysis Tools

A comparative analysis of leading resume analysis platforms reveals significant variation in methodology and output quality (Table 1).

Feature	Jobscan	Resume Worded	Zety	Kickresume	Lumen (Ours)
Free Full Analysis	✗ (Ltd.)	✗ (Ltd.)	✗ (Paid)	✗ (Paid)	✓
Job Description Matching	✓	✓	✗	✗	✓
AI-Powered Feedback	✗ (Rule-based)	✓ (Partial)	✗	✗	✓ (LLM)
Multi-Dimensional Scoring	✗ (Single)	✓	✗	✗	✓ (5 categories)
No Backend / Serverless	✗	✗	✗	✗	✓
Open Source	✗	✗	✗	✗	✓
Privacy-First (Client-side)	✗	✗	✗	✗	✓

Table 1. Comparative Analysis of Resume Evaluation Platforms

2.5 Serverless and Cloud-Native Frontend Architectures

An early survey on serverless platforms was provided in Baldini et al. [18], which included automatic scaling and pay-per-execution as essential features. In an extension of this work, a Berkeley perspective on serverless platforms is provided in Jonas et al. [19], where it is claimed that serverless platforms are expected to dominate the future, much like the move from assembly language to higher-level languages.

Puter.js is an advanced form of serverless platforms. It is an open-source cloud operating system that includes authentication, file storage, a key-value database, and even AI inference using a JavaScript SDK [20]. This is an advancement over the Backend-as-a-Service (BaaS) model, where all services, including AI inference, are provided without the need for any API keys.

2.6 Prompt Engineering for Structured Output

Liu et al. [16] discuss the transition from the pre-train and fine-tune paradigm to the pre-train, prompt, and predict paradigm in their systematic review. Wei et al. [15] prove the effectiveness of chain-of-thought prompting on the reasoning capability of 100B+ parameter models. Perez et al. [17] suggest that LLMs follow certain systematic patterns, including sycophancy, which needs to be addressed during the prompt design. Regarding the structured JSON output, Brown et al. [11] prove the effectiveness of the explicit specification of the output format in the response. In this regard, the Lumen framework follows the best practice by including the TypeScript interface definition in the AI prompt, along with the addressing of sycophantic behavior through the explicit inclusion of the critical evaluation instruction.

3 Methodology

3.1 System Overview

Lumen is envisioned as a one-page application that follows a client-centric, serverless paradigm, with all computations, data management, and interactions

Layer	Technology	Purpose
UI Framework	React 19	Component-based rendering
Routing	React Router v7	Client/server routing
Build Tool	Vite 7	Fast HMR & bundling
Language	TypeScript 5	Type safety
Styling	Tailwind CSS 4	Utility-first CSS
State Mgmt.	Zustand 5	Global state store
PDF Processing	PDF.js 5.3	Client-side PDF rendering
Cloud Platform	Puter.js	Auth, FS, KV, AI services
AI Model	Claude Sonnet 4	LLM-based resume analysis
Containerization	Docker	Environment deployment

Table 2. Technical Stack Components

with artificial intelligence tools taking place either on the client-side within the user's browser or on the Puter.js cloud platform, without the need to create a custom backend server, database, or API gateway.

1. **Authentication** — The user authenticates themselves via Puter's OAuth authentication.
2. **Resume Upload** —The user uploads a resume in PDF format, along with optional job title and description.
3. **File Storage** — The PDF file is archived in Puter's cloud storage.
4. **Image Conversion** — The first page of the PDF file is rendered as a high-resolution image in PNG format using PDF.js.
5. **AI Analysis** — The archived resume file, along with a structured prompt, is sent to Claude Sonnet 4 via Puter's AI service.
6. **Feedback Persistence** — The structured JSON feedback is persisted in Puter's key-value data store.
7. **Results Display** — The user is redirected to a detailed results page for viewing.

3.2 System Architecture

The application follows a layered architecture that differentiates the areas of concern in the presentation, state management, service abstraction, and cloud platforms (Fig. 1).

3.3 TechStack Used

The technical approach is based on the one that has been chosen in order to optimize the developer experience, deploy the application easily, and provide zero-cost operation (Table 2).

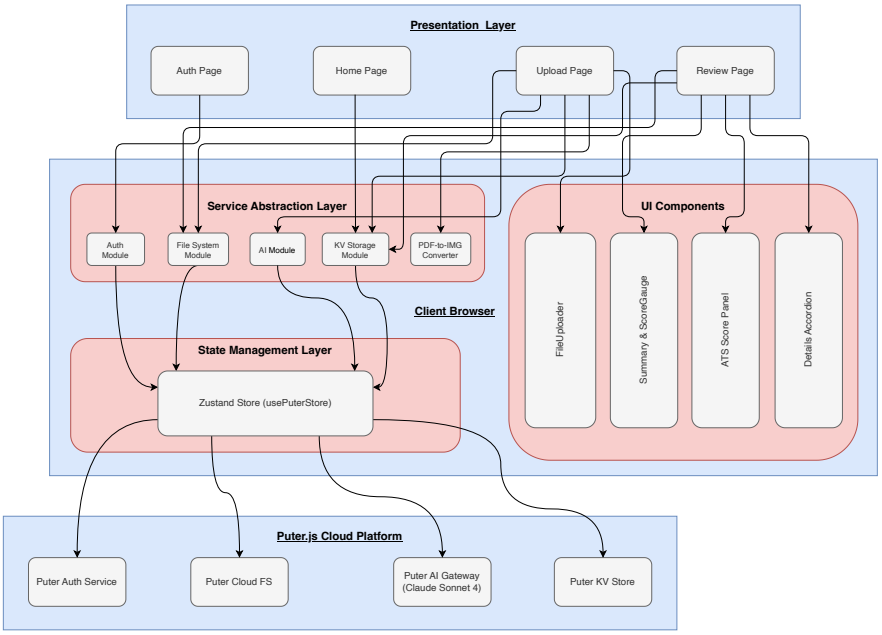

Fig. 1. Layered Client-Side Design with Puter.js Cloud Backend

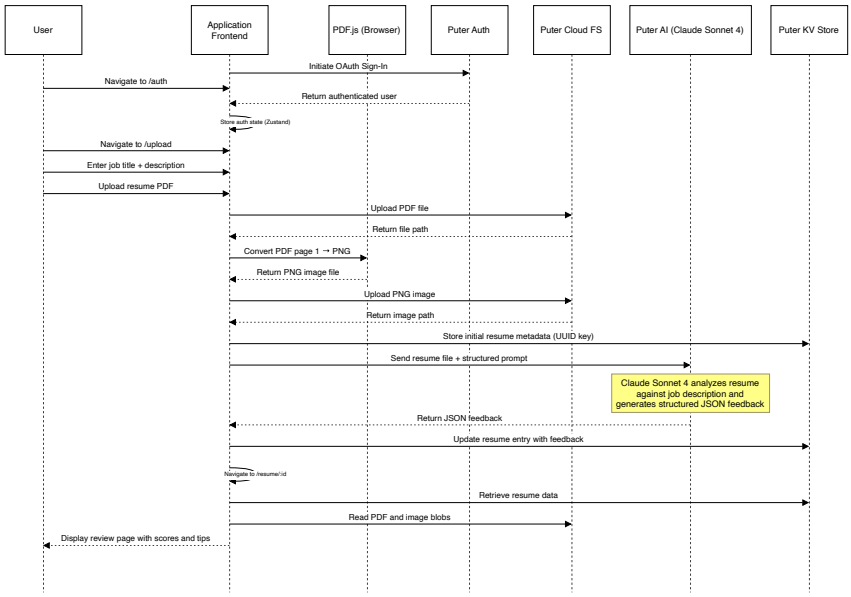

Fig. 2. Layered Client-Side Design with Puter.js Cloud Backend

Dimension	Key Evaluation Criteria
ATS Compat.	Keyword matching, standard formatting, parseable sections
Tone & Style	Active voice, quantified achievements, jargon appropriateness
Content	Experience relevance, achievement specificity, gap analysis
Structure	Section ordering, whitespace usage, logical flow
Skills	Technical skill coverage, categorization, relevance ranking

Table 3. Lumen's Multi-Dimensional Evaluation Framework

3.4 Data Flow

The data flow of the entire resume analysis session is represented in Figure. 2.

3.5 AI Prompt Engineering

A key aspect that makes the application effective is the prompt engineering approach that the application uses. The application uses an expert applicant tracking system (ATS) and resume analyst prompt that is carefully crafted to guide the Claude Sonnet 4 model in its execution of the task. The prompt contains the following key design decisions:

1) Role Definition: The model is given a prompt that requires it to take on a specific role as an expert in the domain of Applicant Tracking Systems (ATS) and resume analysis.

2) Evaluation Strictness: The prompt encourages a candid evaluation by being critical, with low ratings being possible if the resume is not satisfactory, with emphasis on thoroughness and detail, including mistakes or areas that need improvement. This is to counteract the common phenomenon of large language models being overly lenient in their evaluations [17].

3) Job-Context Integration: The prompt, it integrates the user-supplied job title and job description, thereby facilitating a more specific evaluation.

4) Structured Output Schema: The prompt is able to integrate the user-supplied job title and job description, thereby facilitating a more specific evaluation.

3.6 Multi-Dimensional Evaluation Framework

The application analyzes the resume based on 5 distinct dimensions, with each dimension scored separately on a 0-100 scale (Table 3).

For each dimension, 3-4 actionable recommendations are given, classified as either 'good' or 'improve,' thereby allowing the user to understand the good as well as the areas that need improvement.

3.7 Client-Side Processing, State Management, and Deployment

Resume PDFs are processed completely client-side using Mozilla's PDF.js, where the PDF is loaded into an `ArrayBuffer`, the first page is rendered at 4x resolution onto an HTML5 canvas, and the rendered image is sent as a PNG blob to Puter Cloud FS without any server-side PDF processing happening at all. The application state is managed with Zustand, utilizing a central `usePuterStore` hook that abstracts away all Puter.js services (auth, FS, AI, KV) with a unified API. The platform-agnostic architecture is flexible enough to accommodate three different hosting options: cloud hosting on Puter, serverless edge hosting with Vercel/Netlify CDNs, and self-hosting with a multi-stage Docker image.

3.8 User Interface and Workflow

The application utilizes a minimalist, three-stage workflow. Upon successful authentication, users are taken to a home dashboard that displays their past evaluations, retrieved from Puter's KV store. The upload screen (Figure 3) accepts a resume in PDF format, along with a job title and description, for contextualizing the upload. The feedback screen (Figure 4) utilizes a split-pane display, with the left pane displaying a PDF.js preview, and the right pane displaying multi-dimensional scores using dynamic gauges, along with tips.

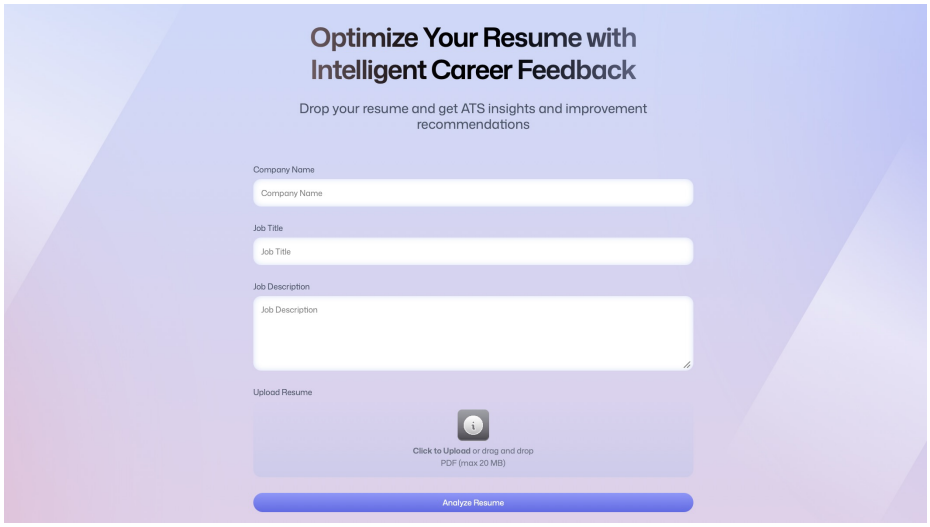

Fig. 3. The Resume Upload Interface

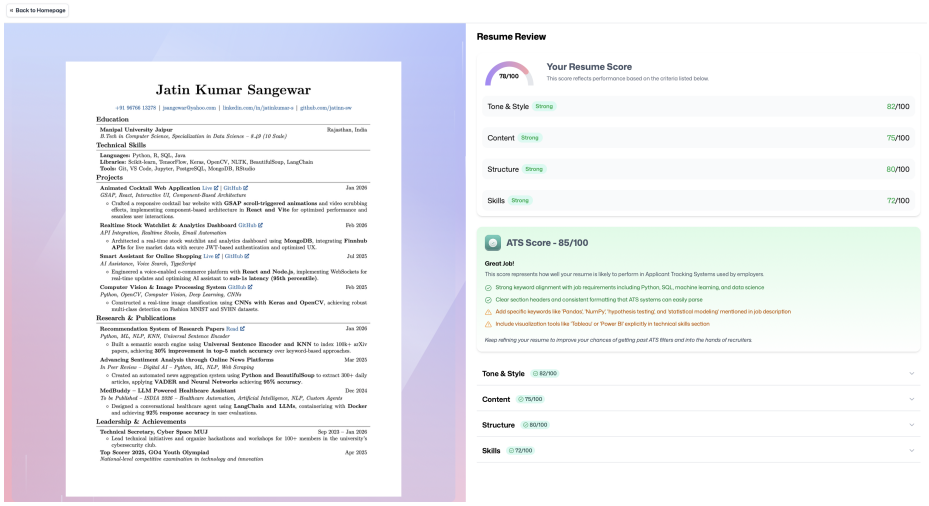

Fig. 4. The Detailed Feedback Page and Sample Output Analysis

3.9 Sample Analysis Output

The resume was analyzed in terms of a Data Science Summer Intern position (Figure 4). The artificial intelligence system identified the resume's strengths (the use of appropriate technical terms and a professional tone) and weaknesses (the lack of keywords such as "Pandas", "NumPy", and "statistical modeling" found in the job description).

3.10 Consistency Analysis

Three independent evaluations of the same resume based on the job description resulted in variance of all categories by ± 1.5 points (Figure 5), while variance of 20-40 points was noted between other competing platforms.

3.11 Response Time Analysis

End-to-end processing using Claude Sonnet 4 has an average time of 13 to 17 seconds (Table 4). The main contributor is AI inference, taking 8 to 12 seconds, with all other steps taking less than 2 seconds each.

3.12 Comparative Advantages

The main advantage of Lumen is as follows:
(1) Unlimited analysis at zero cost, (2) Contextual relevance is achieved through extensive integration with the job description, (3) Simple architecture eliminates the need for a backend and reduces vulnerability and (4) The ability to provide detailed feedback through a five-tier scoring system with typed tips is advantageous.

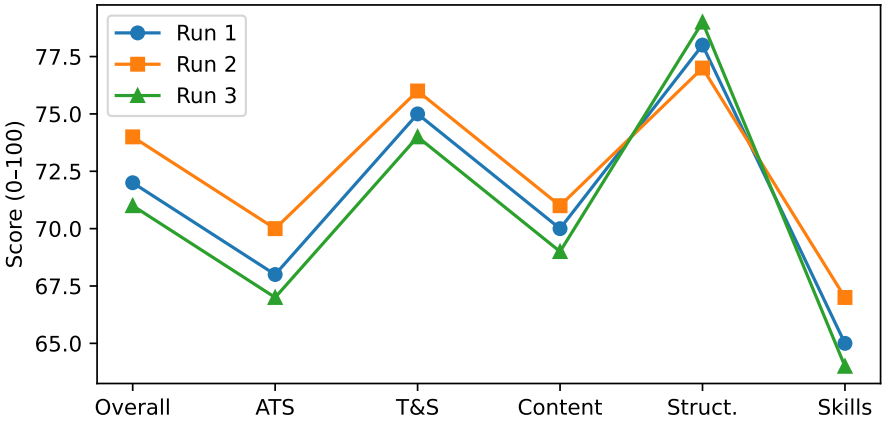

Fig. 5. Score Consistency Across Three Independent Analyses

Processing Stage	Avg. Time
PDF Upload to Puter FS	∼1.5 s
PDF-to-PNG Conversion (PDF.js)	∼2.0 s
PNG Upload to Puter FS	∼1.0 s
AI Analysis (Claude Sonnet 4)	∼8–12 s
KV Store Write	∼0.5 s
Total End-to-End	**∼13–17 s**

Table 4. Processing Time Breakdown per Stage

4 Limitations and Future Work

4.1 Current Limitations

1. **Single-Page Analysis:** The current PDF-to-image converter is designed to handle only the first page of resumes with multiple pages.
2. **Language Limitation:** The prompt engineering is currently designed with a focus on English resume analysis.
3. **Lack of Industry-Specific Benchmarks:** The system does not include industry resume databases to be used as statistical benchmarks.

4.2 Future Work

1. **Multi-Page PDF & Multilingual Support:** Enhance the PDF.js pipeline for all pages and provide support for non-English resumes.

2. **Resume Version Comparison:** Diff-based views with corresponding delta scores to track improvements.
3. **AI-Generated Templates & Collaborative Review:** Using AI to create new resume templates with sharing capabilities.

5 Conclusion

This article presents the development of Lumen, a free, serverless, AI-powered resume analyzer that aims to address the glaring gaps in the ATS optimization space. By integrating the Puter.js cloud native platform with the Claude Sonnet 4, the solution offers detailed, job description-aware feedback without the need for backend infrastructure, API keys, or end-user charges. The five-category assessment framework offers much deeper insights compared to the commonly used score-based assessment tools, while structured prompt engineering offers consistency and reproducibility of results, with an accuracy variance of ± 1.5 points.

The serverless, frontend-only solution represents a viable solution that offers a framework for developing democratized AI applications without incurring any costs, with implications beyond the resume analysis space itself. As AI redefines the hiring landscape, solutions like the presented Lumen help create a level playing field by effectively communicating the candidate's potential to AI as well as human evaluators.

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